Nakshatra Kundlas

Full-Stack, Systems & AI Engineer
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Summary

Full-stack, systems and AI engineer with hands-on production experience shipping backend features, debugging live incidents, running QA/reliability programs and operating self-hosted infrastructure. Experienced across FastAPI, React/Next.js, TypeScript, PostgreSQL, Redis/Celery, Docker, Tauri/Rust, CI/CD and applied ML, with a strong focus on reliable, security-conscious systems and practical AI products.

Technical Skills

Languages	Python, TypeScript, JavaScript, Rust, Java, C/C++, SQL
Backend	FastAPI, Node.js, Express, Spring Boot, REST APIs, WebSockets, SQLAlchemy, Pydantic, Celery, ARQ, Convex
Frontend	React, Next.js, Tailwind CSS, React Flow, responsive UI engineering
AI / Data	PyTorch, TensorFlow, scikit-learn, Hugging Face, Pydantic AI, LiteLLM, NumPy, Pandas, Vertex AI
Systems / Infra	Linux, Docker, Tauri, QEMU/KVM, GitHub Actions, self-hosted runners, Coolify, PostgreSQL, Redis, object storage, Tailscale, Cloudflare Tunnels
Testing / Reliability	pytest, unit/integration/security testing, Playwright, migration testing, observability, production debugging and release verification

Experience

Backend and Systems Engineer
Apr 2026 - Present

Analytics Depot - Freelance — *Remote - New York, USA*

- Ship backend, reliability and AI-analytics work into a production analytics platform across document/data ingestion, connectors, scheduled jobs, observability/evaluation and CI/CD.
- Troubleshoot live failures from server logs, PostgreSQL/Supabase state, Redis/Celery workers and API traces; reproduced and fixed user-facing ingestion, chunking/vector retrieval, connector-sync and upload/chat-state bugs.
- Implemented security and reliability remediations for sandbox/SQL execution, OAuth credentials/state, tenant/cache isolation, task authorization, email sanitization, worker isolation and idempotent scheduled delivery.
- Contributed to agent observability/evaluation work and performed extensive QA/release verification, including a 45-case Unicorn Companies benchmark, production-shaped migration probes, regression gates and manual live smoke testing.

Software Engineering Intern
Jan 2026 - May 2026

Seven Oceans Holdings — *Remote - Singapore*

- Built and deployed a production support/CRM platform for Voyage Management Enterprise Software, replacing legacy ticket workflows and centralizing operational requests.
- Developed a type-safe React 18/TypeScript frontend and FastAPI/PostgreSQL backend with ticket, customer and vendor workflow surfaces.
- Implemented AI-assisted request routing and vendor workflow automation using LiteLLM, Pydantic AI and multi-tool integrations.
- Used Redis and ARQ for asynchronous long-running resolution workflows and resilient background processing.

Open Source Contributor & Project Maintainer
2025 - Present

GitHub — *Public + upstream projects*

- Recorded 1.2K+ GitHub contributions in 2026 across private production engineering, public projects and upstream open source.
- Landed 5 merged upstream PRs across Tracer Cloud/OpenSRE and Pulse, including tested file-sync transfer telemetry and Tauri/Rust + TypeScript/Svelte privacy controls.

Selected Projects

Homelab
Self-Hosted Production Infrastructure & CI/CD Platform

Linux, Docker, Coolify, GitHub Actions, PostgreSQL, object storage, Tailscale, Cloudflare Tunnels

- Converted a minimal Linux installation into a 24/7 bare-metal server platform running production websites, internal apps and isolated container workloads alongside databases, object storage and automation services.
- Built a self-hosted deployment stack with Coolify and a GitHub Actions runner for CI, container builds and automated delivery behind reverse-proxy and custom-domain routing.
- Operate home automation, workflow engines, Nextcloud, Plex and private download/media services, with Cloudflare Tunnels and Tailscale providing secure remote access while reducing reliance on managed cloud compute.

Agent OS
Self-Hosted Reliability Control Plane for AI & Automation

Node.js, PostgreSQL, Docker Compose, AI/API workflows, payment & wallet controls

- Built a self-hosted control plane where PostgreSQL is authoritative for tasks, approvals, external effects, payments and audit history; durable idempotency/state machines make retries and worker restarts recoverable.
- Enforced pause/kill/spend boundaries in the service layer, signature-verified and deduplicated PayPal webhooks, approval-gated wallet intents and redacted outbox notifications with automated restart/recovery tests.

SandLabX *Network Virtualization and Lab Automation Platform*

Next.js, React Flow, Node.js, QEMU/KVM, Docker, PostgreSQL, Apache Guacamole

- Architected a self-hosted network-lab control plane using QEMU/KVM, QCOW2 copy-on-write overlays, PostgreSQL-backed state and browser VNC/serial access through Apache Guacamole.
- Built managed image pipelines and versioned Lab Capsules with checksum validation, atomic publishing, deterministic network plans, RBAC, rate limiting, audit logging and typed scenario verification.

VibeVoice *Local-First Cross-Platform Voice Input*

Tauri, Rust, TypeScript, whisper.cpp, GitHub Actions

- Built a native dictation app for Linux, Windows and macOS that records audio, transcribes locally with whisper.cpp, cleans developer terminology and inserts text into the focused application.
- Designed privacy-aware transcript/history handling, diagnostics and temporary-audio cleanup, then shipped cross-platform installers, GitHub release automation and in-app update support.

FlashRL *Deterministic Reinforcement Learning Research Benchmark*

Python, PyTorch, Gymnasium, DQN, experiment tracking, GitHub Actions

- Built a deterministic Dino reinforcement-learning benchmark with exact frame budgets, resumable experiments, versioned artifacts and held-out multi-seed evaluation for reproducible research.
- Implemented configurable vanilla, Double and Dueling Double DQN with prioritized replay and N-step returns; trained a five-seed 120K-frame benchmark and evaluated each policy on 100 unseen episode seeds.
- Shipped a live policy laboratory exposing Q-values, actions, rewards and checkpoint identity, with deterministic replay/GIF generation, CI and publication-ready reports.

Additional projects

- **mdview** - Local-first Tauri Markdown desktop app with sanitized rendering, Mermaid, multi-mode editing and native AppImage/DEB/RPM/NSIS/DMG release pipelines.
- **CB Connect** - Real-time consent-first couples health app with Convex/Clerk, revocable partner permissions, raw-body webhook verification and timezone-aware health writes.
- **Dayframe** - Mobile-first privacy-focused sleep, movement and food journal with Apple Health Shortcut import, local IndexedDB storage and ephemeral meal-image analysis.

Education

Bachelor of Technology in Artificial Intelligence and Data Science 2022 - 2026

Shiv Nadar University, Chennai *Chennai, India*

Certifications & Training

- Elements of AI: Introduction to AI, University of Helsinki (Feb 2026) - Credential ID: voj58ocgjjj
- OpenAPI Fundamentals (LFEL1011), The Linux Foundation (Jan 2026) - Credential ID: LF-fq040jy0s2
- 5-Day AI Agents Intensive Course with Google, Kaggle (Dec 2025)
- Prompt Design in Vertex AI Skill Badge, Google (Dec 2025)
- Machine Learning with Python, freeCodeCamp (Dec 2025) - Credential ID: zburgers-mlwp
- PyTorch and Deep Learning for Decision Makers (LFS116), The Linux Foundation (Dec 2025) - Credential ID: LF-d5jxp5xcoc
- Hewlett Packard Enterprise Software Engineering Job Simulation, Forage (Nov 2025) - Credential ID: ynMexduPZbh6mkC5d
- NPTEL: Blockchain and its Applications; Responsible and Safe AI Systems; Patent Law for Engineers and Scientists